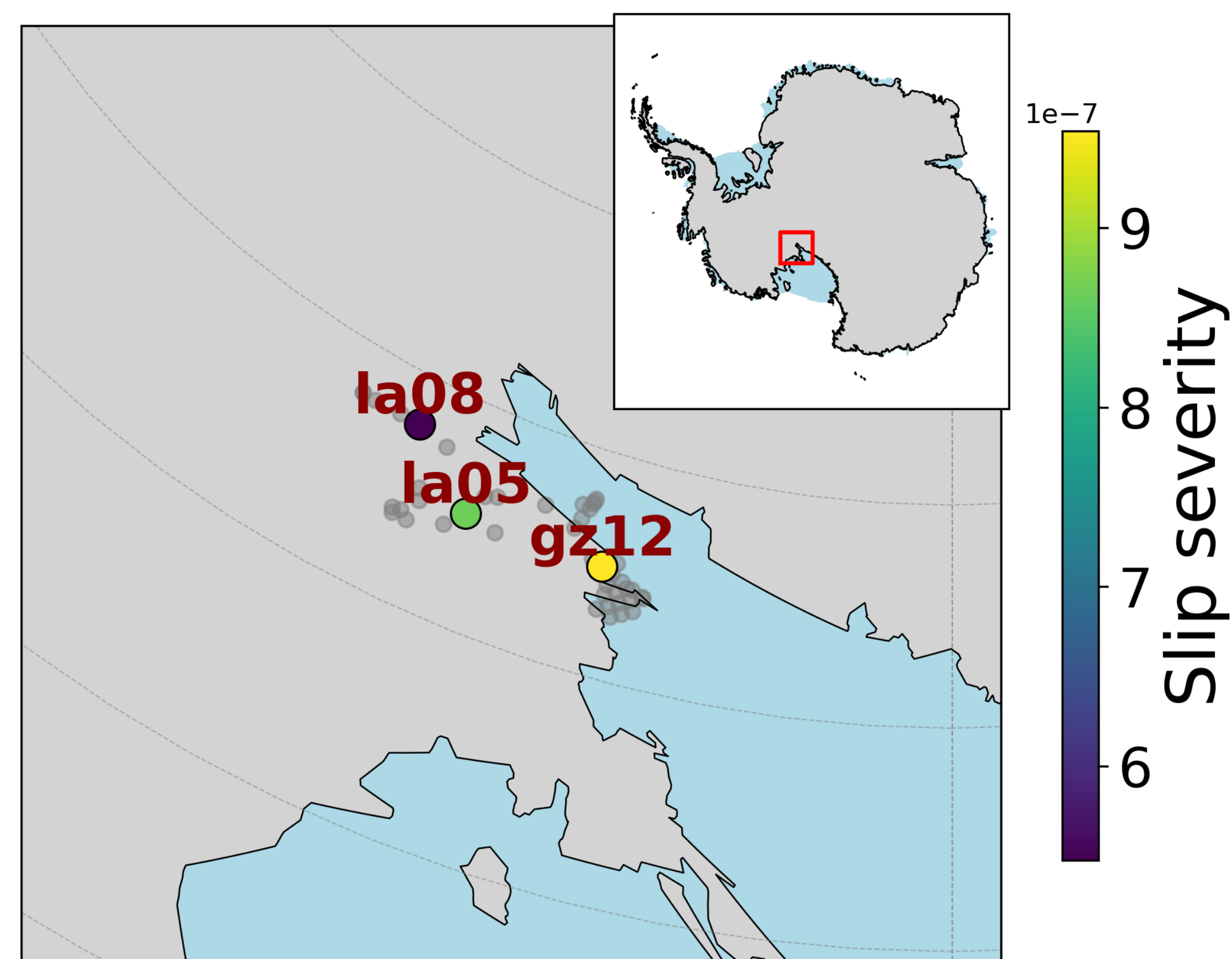
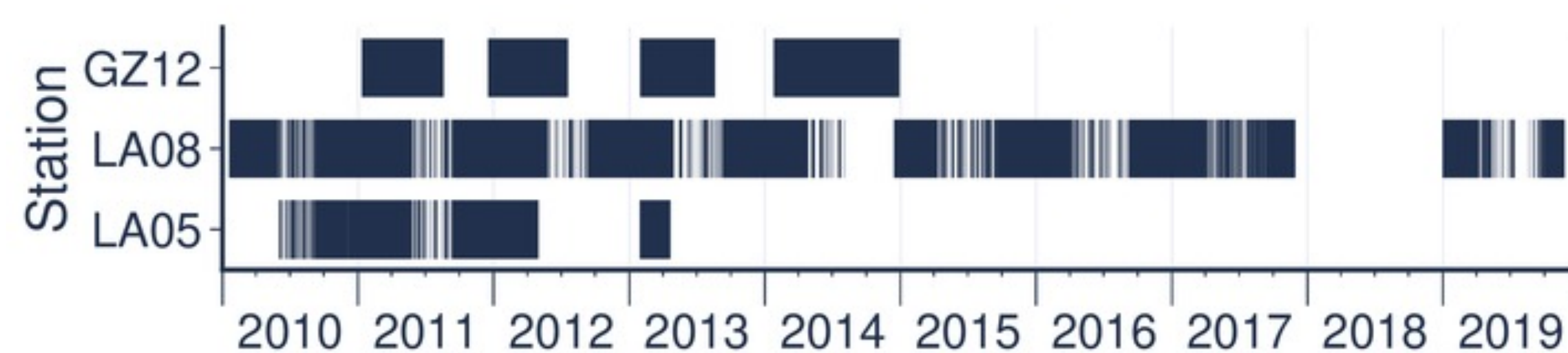


Whillans Ice Stream

Whillans Ice Stream, a major outlet of the West Antarctic Ice Sheet flowing into the Ross Ice Shelf, provides critical insights into ice-ocean interactions, subglacial processes, and ice-sheet stability.

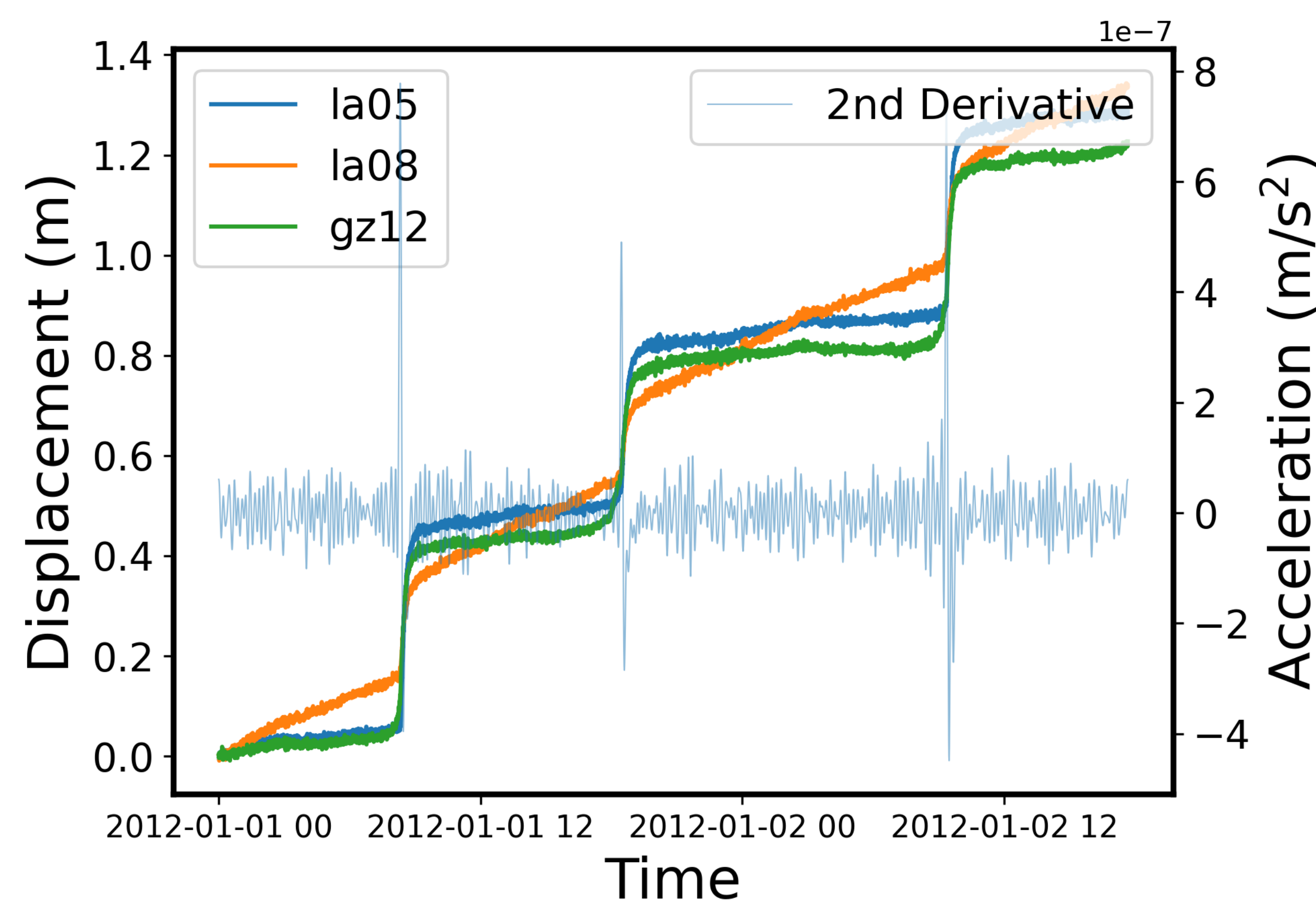


Global Navigation Satellite System (GNSS) data has been collected from 2008-2019, stations recording position every 15-30 seconds.



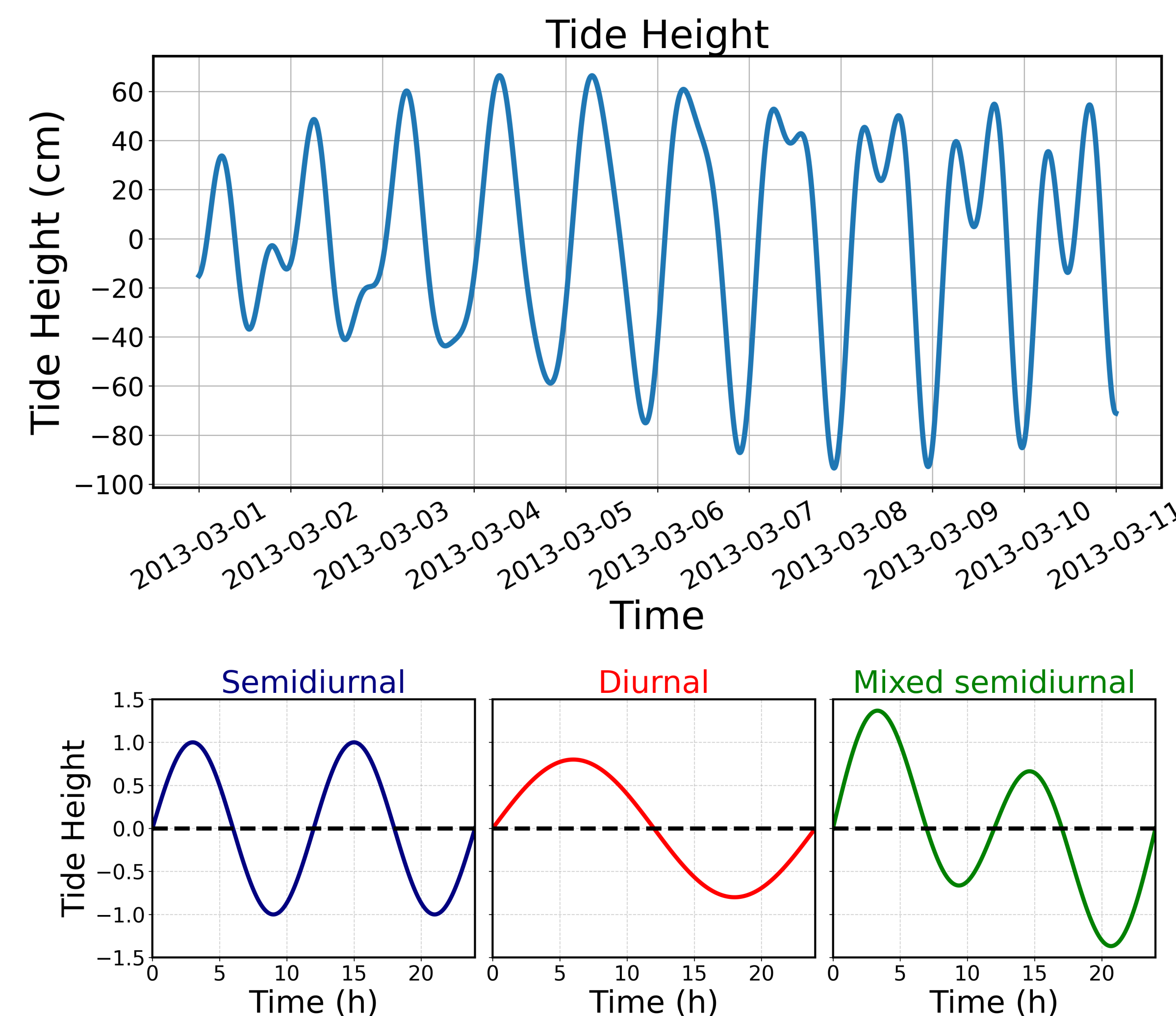
Stick-Slip Motion

Whillans glacial movement follows a stick-slip pattern, with one to two slip events per day, driven by tidal forcing, basal friction, and other factors. An event catalog was produced by a residual based detection algorithm.



Data Features and Trends

An essential part of machine learning is the creation of features (predictors) for the model. The models incorporate multiple tidal features, including tide height, form factor, tidal derivative, and inter-event diurnal/semi-diurnal amplitudes.



Other extracted features include slip severity, pre-slip area, time since the previous event, inter-event R^2 , displacement, and slope.

Together, these metrics capture both the magnitude of slip and the timing between events, providing a fuller picture of the stick-slip cycle.

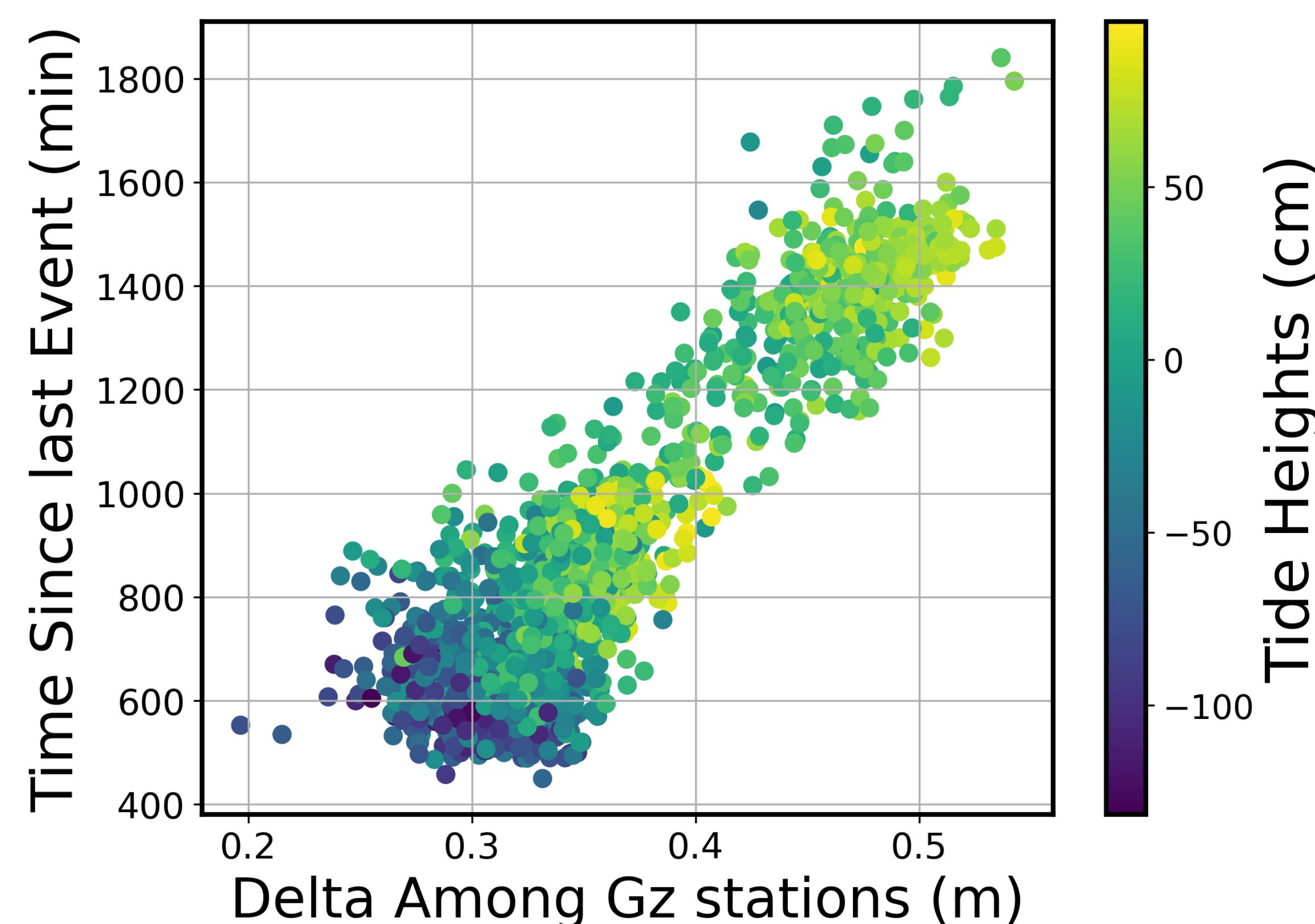
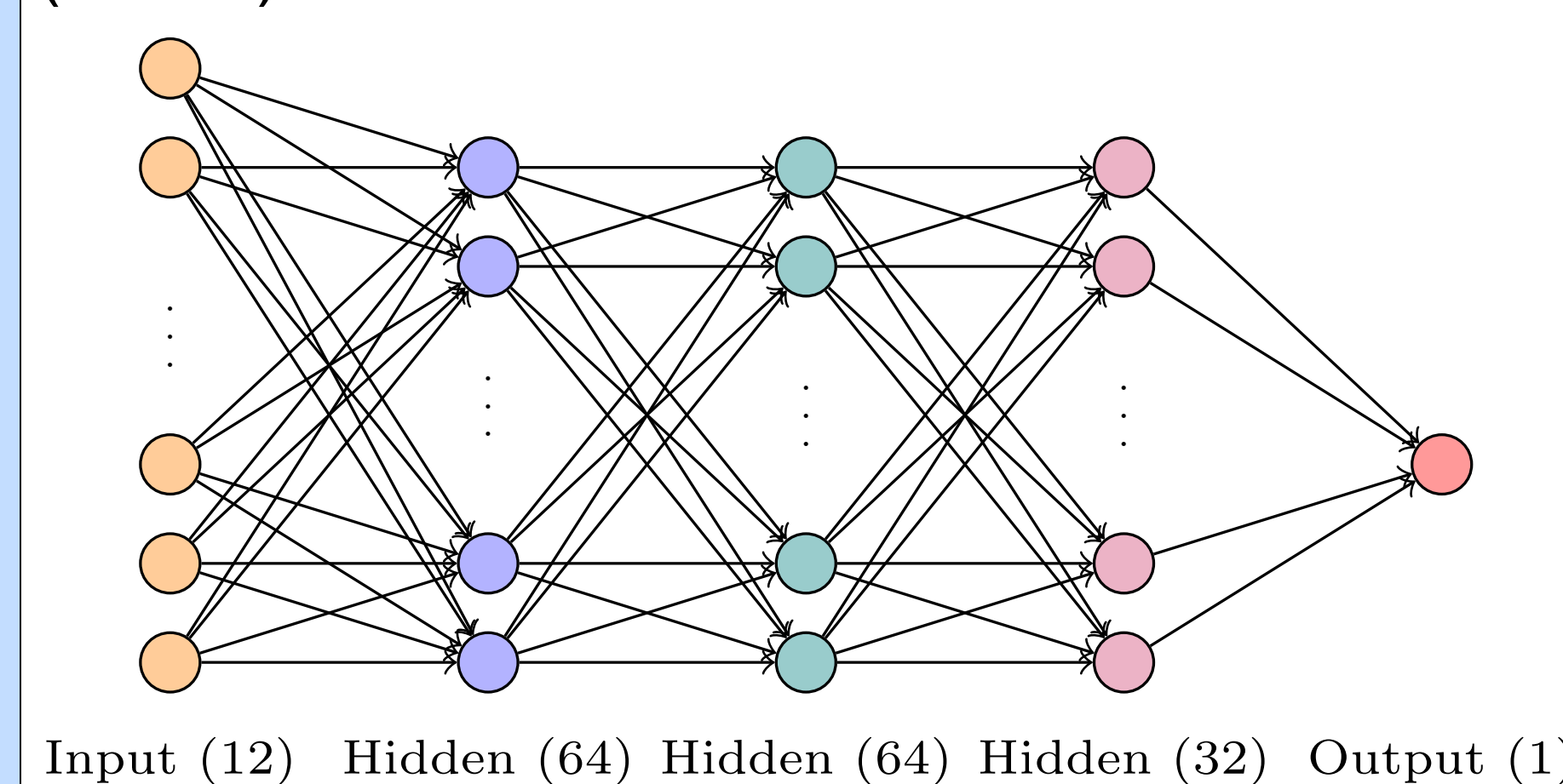


Figure shows the linear trend between inter event time and slip size. Clustering is also apparent with higher tide events generally slipping more than lower tide events.

Predicting Slip Magnitude with Machine Learning

A key goal of this research was to predict important ice-stream metrics using engineered tidal and displacement features. By pairing these features with multiple models, we achieved strong performance and identified which factors most strongly influence slip behavior. Time-series models, including Long Short-Term Memory (LSTM) networks were tested but showed limited effectiveness.

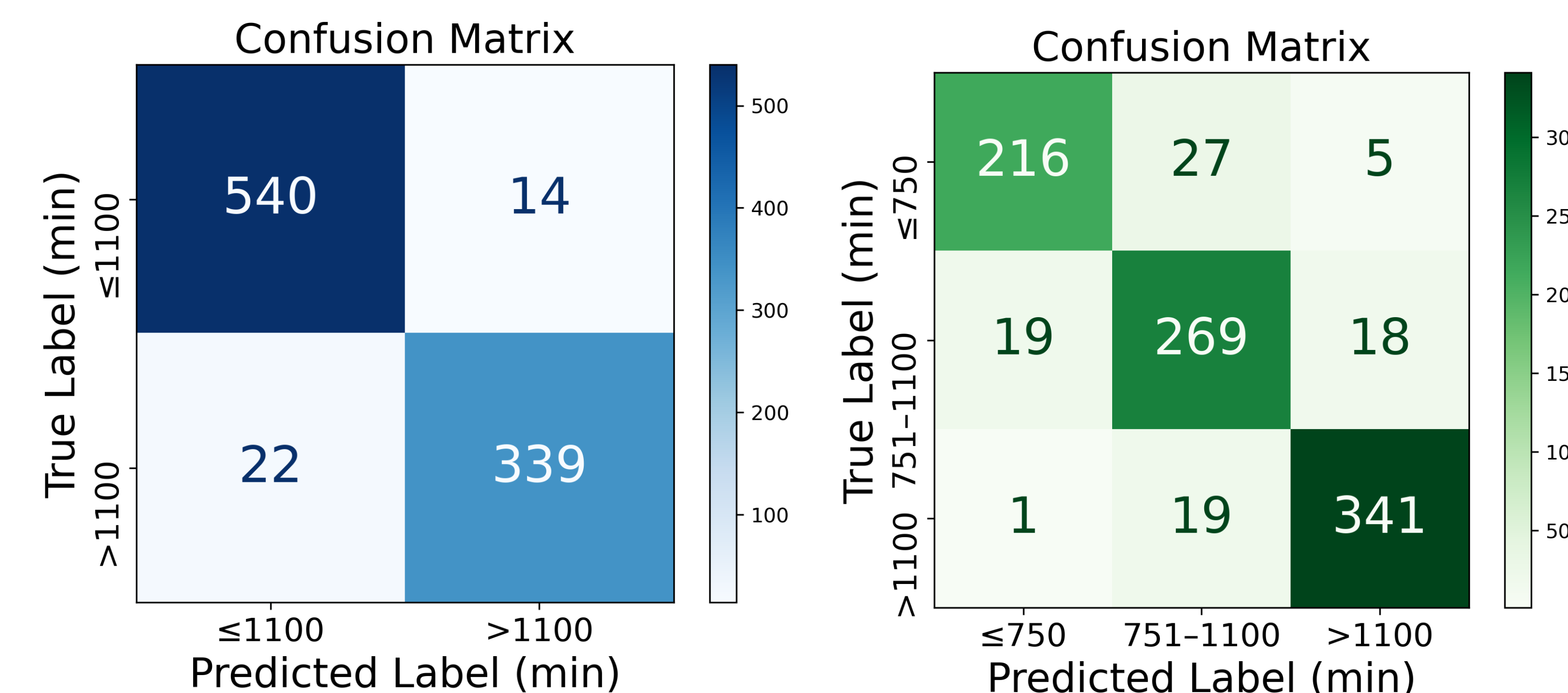


Model	R^2
Linear Regression	0.823
Neural Net	0.943
LSTM (Neural Net)	0.625
XGBoost	0.916

Machine Learning Predictions of Event Timing

Slip event timing proved highly complex, and regression models failed to capture the variance effectively. We therefore applied classification models, which successfully identified the tidal-modulated patterns controlling event timing. The highest performing model being Support Vector Machine (SVM).

Model	Target split	Accuracy
SVM	0-1100, 1100+ minutes	0.903
SVM	0-750, 750-1100, 1100+ minutes	0.961



Summary and Implications

Exploratory data analysis and standard data science methods revealed key patterns in ice-stream motion. Machine learning models predicted many aspects of this behavior, with the most relevant being slip size and time until the next event. Much of Antarctica's geophysics remains unknown, making research on its ice dynamics essential for anticipating future sea-level rise and global impacts. Future work could involve seismic analysis, investigating data imputation strategies, and the exploration of advanced machine learning approaches such as transformer architectures.

Acknowledgements:

This work was funded by Mines SURF and NSF award 2145407